



澳門大學
UNIVERSIDADE DE MACAU
UNIVERSITY OF MACAU

Faculty of Science and Technology

Taste of Macau: A Visual Food Finder

Team Project Proposal

By

Team 04

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1. Project Title

Project Topic Name: Taste of Macau: A Visual Food Finder

Precisely Written Question: How can an interactive, visually driven web application be designed to effectively catalogue and present information about Macau's diverse food culture, allowing users to browse and discover dishes based on categories and location, while providing complete user account management (signup/email verification/login/password reset), a personal user dashboard, and e-commerce style features (search history, browsing history, and favorites)?

2. Project Purpose

2.1. Outcomes

This project directly connects to the core learning outcomes of the CISC3003 Web Programming course by requiring the practical application of front-end, back-end, and database development principles. According to MDN Web Docs (2024), modern web development demands proficiency in combining HTML, CSS, and JavaScript to create functional and interactive platforms, which is precisely what this project entails. For our majors in Computer Science, this project demonstrates the ability to build a full-featured, user-centric web application that manages user accounts, manages data persistence, provides a personal user dashboard, and delivers interactive services. The application features a trilingual architecture supporting Chinese, English, and Portuguese — Macau's three official languages — with every food entry, restaurant description, and UI label existing in all three languages. In the context of future study or careers in web development, this project serves as a substantial portfolio piece. As W3Schools (2024) emphasizes, hands-on project experience is crucial for demonstrating employability skills in the competitive web development industry. GenAI tools will also be explored to assist with content generation and code debugging throughout the development process.

2.2. Skills

This project will help us practice the following essential skills.

First, **Responsive Web Design** using CSS with Flexbox and Grid to ensure the application is usable and visually appealing on all devices. Based on the principles outlined by Google's Web Fundamentals (2024), responsive design is no longer optional but a necessity in modern web development.

Second, **Interactive Front-End Programming** by writing JavaScript to manage user events and dynamically update content. MDN Web Docs (2024) highlights that DOM manipulation is a fundamental skill for creating dynamic user experiences.

Third, **Third-Party API Integration** by incorporating a mapping library (Leaflet) with custom restaurant clustering built on Leaflet.

Fourth, **Data Management** through structuring and manipulating complex data in JavaScript arrays and objects, as well as using databases for persistent storage.

Fifth, **Basic PHP Programming** to manage server-side tasks including user authentication, form processing, and database interactions. As we have recently learned in class, PHP allows us to add complete back-end functionality.

Seventh, **User Dashboard Development** – Designing and implementing a personal dashboard where logged-in users can manage their activities, including viewing search history and browsing history in one centralized interface.

Eighth, **Search Functionality** – Implementing a search feature that allows users to find food items by name, category, or location, with search queries saved to each user's dashboard history.

Ninth, **Favorites System** – Designing and implementing a favorites collection where users can save dishes. Guest favorites are stored in LocalStorage and synced after login.

Tenth, **Web Hosting on PHP** – Supporting Platforms to deploy and share our project online. We will deploy on our own Apache server rather than Wasmer, giving us full control over mod_rewrite, SSL configuration, and SQLite WAL-mode settings.

Finally, **Visual Communication** by curating and presenting images and information effectively. According to the Macao Government Tourism Office (2024), visual storytelling is key to promoting Macau's unique gastronomic culture.

2.3. Knowledge

This project will help us become familiar with the following important content knowledge in this discipline.

Document Object Model (DOM) Manipulation will provide a deeper understanding of how to programmatically access and change HTML elements and their attributes. Eloquent JavaScript (Haverbeke, 2024) describes the DOM as the bridge between JavaScript and the content of a webpage.

Asynchronous JavaScript concepts will be explored if implementing advanced features like fetching food data from a JSON file, where we will learn the basics of asynchronous operations using `fetch()` and Promises as documented by MDN Web Docs (2024).

Geospatial Data in Web Apps knowledge will be gained through understanding how to work with coordinates and display points of interest on maps, following Leaflet's tutorials (2024) on efficiently managing location-based data.

Basic PHP Syntax and Usage will be applied to create server-side features such as authentication, form processing, and database connections, reinforcing what we have recently learned about server-side scripting with PHP.

User Authentication Concepts will be understood how to implement secure user registration, login sessions, email verification tokens, and password reset workflows using PHP and basic session management.

Email Integration will be learnt how to send verification emails and password reset links using PHP's mail functions or SMTP.

User Dashboard Design will be understood how to create a personalized user interface that aggregates multiple services (search history, browsing history) into a single, coherent dashboard view.

Search Algorithms will be applied basic string matching or filtering techniques to provide real-time or form-submitted search across the food database.

Session Management will be understood how to maintain user login state across pages using PHP sessions.

Database Design how to design a simple relational database (e.g., MySQL or SQLite) to store user accounts, user search history, and browsing history.

Deployment on PHP-Supporting Hosting will provide practical experience in publishing a web project to a live URL, including file structure management, database setup, and basic version control with Git.

Marker Clustering Techniques – Understanding how to implement marker clustering (e.g., using Leaflet.markercluster plugin) to handle overlapping map markers at lower zoom levels, and designing bi-directional linking logic between food cards and clustered markers.

Guest User Data Management – Learning how to handle browsing and search history for non-logged-in users using browser LocalStorage, and implementing merge logic when a guest user logs in.

Trilingual Application Architecture – Understanding how to implement a full i18n system with database column suffixes (name_en, name_zh, name_pt), server-side and client-side language switching, and AJAX-persisted locale preferences.

Finally, **Information Architecture** principles will be applied to structure content for optimal discoverability, as the Macao Government Tourism Office (2024) notes that Macau's UNESCO creative city designation in gastronomy requires effective digital presentation of culinary heritage.

3. Project Tasks

To complete this project successfully within four weeks, a systematic approach will be followed, guided by the development methodology suggested by MDN Web Docs (2024). With six team members, we will divide responsibilities across front-end, back-end, database, mapping, and deployment. Weekly milestones are defined below.

3.1. Week 1 – Foundation & User System

Phase 1: Planning and Research (Days 1–2)

We will research and curate a list of 45–50 iconic Macau dishes and foods. According to the Macao Government Tourism Office (2024), Macau's gastronomy reflects over 400 years of cultural fusion between Portuguese and Chinese culinary traditions.

Food Selection Criteria: Our selection is based on three weighted dimensions:

- **Cultural Representativeness (40%):** Prioritizing iconic foods that represent Macau's Portuguese-Chinese fusion heritage, such as Portuguese Egg Tart, Pork Chop Bun, Minchee, and African Chicken, as recognized by MGTO.
- **Geographical Distribution (30%):** Deliberately covering Macau Peninsula, Taipa, and Coloane to ensure map diversity and usefulness across all regions.
- **Popularity & User Recognition (30%):** Referring to MGTO official data and major food platforms (e.g., OpenRice, Dianping) for ratings and review counts.

We will select foods that best represent this unique heritage, organized into categories such as:

- Street Snacks (e.g., Pork Chop Bun, Egg Waffles, Fish Balls)
- Portuguese & Macanese Dishes (e.g., Portuguese Egg Tart, Minchee, African Chicken, Bacalhau)
- Desserts & Bakeries (e.g., Almond Cookie, Serradura, Egg Roll)
- Main Dishes (e.g., Curry Crab, Clam in Black Bean Sauce)

Additionally, we will curate a dataset of 16 Macau restaurants, each with trilingual descriptions, address, contact information, opening hours, price range, Google and Amap ratings, and geo-coordinates.

For each selected food item, we will gather high-quality images, write concise descriptions, and document their common locations throughout Macau.

Location Data Accuracy Assurance: We will implement a three-layer verification mechanism:

1. **Official Sources First:** Prioritizing addresses and coordinates from MGTO and Municipal Affairs Bureau.
2. **Cross-Validation:** Cross-referencing Google Maps, OpenStreetMap, and Dianping/OpenRice; discrepancies resolved by prioritizing official data or on-site verification.
3. **Manual Spot-Checking:** Physically verifying at least 5 representative locations (e.g., Ruins of St. Paul's area, Taipa Village) due to the manageable dataset size of 20 items.

The deliverable for this phase will be a structured dataset (JavaScript array of objects, later imported into a database) with all compiled food information.

Data Update & Maintenance Strategy:

- **At submission:** All venue statuses will be manually verified (via phone, Google Maps status, or on-site check) before final submission.
- **Feedback mechanism:** A 'Report Location Issue' button in the Dashboard allows users to report closed venues or suggest new locations.
- **Admin panel (internal):** A simple administrative interface for team members to mark venues as 'closed' or update status without code changes.
- **Data freshness label:** Each food card will display the date when information was last verified, with a disclaimer to check official sources.
- **Future extension:** For long-term maintenance, we may integrate Google Places API for automatic periodic synchronization.

Phase 2: Wireframing and Design (Days 2–3)

We will sketch the layout of the main webpage and the user dashboard. Based on responsive design principles from Google (2024), we will plan a mobile-first approach that scales up gracefully to tablet and desktop views. The sketch will include:

- Main page: header (with login/registration links and user status), category filter buttons, search bar, food card grid, embedded map area.
- User Dashboard: a dedicated page/section where logged-in users can view their search history and browsing history.

Phase 3: User Authentication System (Days 4–7)

We will implement a complete user authentication system using PHP:

- User signup with email verification (sending a unique link via PHP mail).
- User login with session management and password verification (using PHP's `password_hash()` and `password_verify()`).
- Password reset flow (forgot password → email reset link → set new password).
- CSRF protection across both SSR form submissions and AJAX API calls.

A simple database (MySQL or SQLite) will store user accounts with fields: id, username, email, hashed password, email_verified flag, verification token, reset token, and reset token expiry.

Week 1 Deliverable: Working user authentication system (signup/email verification/login/password reset) with SQLite database schema (core relational tables with foreign keys and cascading deletes).

3.2. Week 2 – Front-End & Core Features

Phase 4: HTML and CSS Structure Implementation (Days 8–10)

We will write the basic HTML structure and implement the core styling. Following W3Schools' (2024) best practices for semantic HTML and modern CSS layouts, we will create the responsive grid system for the food cards using Flexbox and CSS Grid, style the interactive filter buttons and search bar, and reserve a container for the map integration. Our CSS follows a design-token system inspired by Macau's visual identity — Portuguese street-name tile aesthetics with Chinese red, Portuguese green, and egg tart gold as our core palette. The deliverables will be the foundational `index.html` and `style.css` files.

Phase 5: JavaScript Development – Rendering & Filtering (Days 10–12)

We will create the JavaScript dataset (or fetch from a JSON file/API) containing all food data and write the functions dynamically rendering HTML cards in the grid. As explained in Haverbeke (2024), using array methods like `filter()` and `map()` will enable efficient data manipulation. We will implement category filter logic so that clicking a category button instantly shows only food items belonging to that category, with a "Show All" button restoring the complete list. We will also build the restaurant catalog with service modes (All / Must-Eat / High-Rated / Nearby).

Phase 6: Search Feature Implementation (Days 12–14)

We will implement a search bar that allows users to search for food items by name, category, or location. Search results will dynamically update the food card grid using JavaScript-based filtering.

Favorites System:

- Users can save dishes to a personal favorites collection.
- Guest users save favorites to `localStorage`.
- Upon login, favorites sync to the server with deduplication using `INSERT OR IGNORE` semantics with a unique constraint on `(user_id, food_id)`.

Guest User Search & Browsing History Handling:

- **For guest (non-logged-in) users:** Search queries and browsing history will be stored temporarily in browser LocalStorage. Data remains on the client's side and is not written to the backend database.
- **Upon login:** The system automatically syncs to guest favorites after login.
- **Privacy protection:** No personal identifiable information is collected from guest users. All local data stays on the user's device.
- **For logged-in users:** All search queries are saved to the database and displayed in the Dashboard under 'Search History'.

Week 2 Deliverable: A fully responsive, trilingual front-end with category filters, restaurant catalog, real-time search, and favorites system (with LocalStorage for guests and database for registered users).

3.3. Week 3 – Map Integration & E-Commerce Service

Phase 7: Map Integration and Interactivity (Days 15–18)

We will integrate the open-source Leaflet.js mapping library following their quick start guide (Leaflet, 2024).

Dual-Mode Map Rendering:

- **Food markers:** Individual color-coded markers (by category) with category letters displayed on markers.
- **Restaurant clusters:** Proximity-based clustering at low zoom levels using custom restaurant clustering built on Leaflet.

Marker Clustering Implementation:

- We will implement **marker clustering** using custom restaurant clustering built on Leaflet to handle overlapping markers when users zoom out to the Macau Peninsula level.

- **Card-to-marker linking with clusters:**
 - When a user clicks on a food card, the system checks whether the corresponding marker is currently hidden inside a cluster.
 - If the marker is clustered, the map automatically **zooms in to the level where the cluster expands** and then highlights the individual marker (e.g., change marker icon color or open a popup).
 - If the marker is already visible, the map centers on it and highlights it directly.
 - Special handling for the edge case where a restaurant has multiple signature dishes at the same coordinates.
- **User experience:** A smooth zoom animation clearly shows the transition from cluster to individual marker.

We will add interactive markers for each food location and implement bi-directional interaction, ensuring that when a user clicks on a food card, the corresponding map marker highlights or the map centers on that location. The deliverable will be a fully functional, interactive map embedded in the page with linked markers and clustering support.

Phase 8: User Dashboard Implementation (Days 18–21)

We will design and implement a User Dashboard – a dedicated page accessible only to logged-in users. The dashboard will integrate three service modules:

1. **Search History Service** – Displays a list of the user's past search queries (saved from Phase 6), with the ability to re-run a search by clicking on a previous query.
2. **Browsing/Viewing History Service** – Tracks which food cards the user has clicked on or viewed, stores this in the database, and displays a "Recently Viewed" section in the dashboard.
3. **Favorites Collection** – Displays dishes the user has saved, accessible from the dashboard.

Review System:

- Create and delete reviews with 1-5 star ratings; update API supported.
- Duplicate prevention (one review per user per restaurant).
- Automatic restaurant rating aggregation (average rating and distribution).
- Reviews linked to restaurants and optionally to specific foods.

Restaurant Detail Pages: We will build individual restaurant detail pages, each showcasing signature dishes, user reviews with star ratings and distribution, tag-based attributes, and embedded maps.

Admin Panel: We will develop an admin panel for restaurant and food CRUD management, food-to-restaurant relationship linking, and feedback report triage (status workflow: new → reviewing → resolved).

The dashboard will also display user profile information (username, email, account creation date), a "Report Location Issue" button for user feedback, and a logout button.

Week 3 Deliverable: A fully interactive map with dual-mode rendering, complete User Dashboard with search history, browsing history, and favorites, full review system, restaurant detail pages, and admin panel.

3.4. Week 4 – Back-End Integration, Deployment & Testing

Phase 9: Back-End Integration & PHP Form Processing (Days 22–24)

We will connect all front-end components to the PHP back-end. This includes:

- Processing login and registration form submissions.
- Handling email verification and password reset requests.
- Saving search queries to the database for logged-in users.
- Saving browsing history (viewed food cards) to the database.
- Processing user reviews (create, edit, delete) with automatic rating recalculation.
- Managing admin panel CRUD operations.

- Implementing session checks to conditionally display dashboard content and user-specific data.
- Trilingual content delivery with locale persistence via AJAX.

The deliverable will be a fully integrated web application where PHP manages all server-side logic.

Phase 10: Deployment on Apache Server (Days 24–26)

We will deploy our project on our own Apache server at <https://food.jackie-macau.top> rather than Github, giving us full control over the environment.

We will:

- Upload all project files (HTML, CSS, JS, PHP) to the Apache server.
- Configure `mod_rewrite` via `.htaccess` for clean URLs.
- Set up SSL for secure HTTPS connections.
- Configure SQLite with WAL journal mode for better concurrency.
- Set appropriate file permissions for SQLite database and storage directories.
- Test that all features work correctly in the production environment.
- Obtain a public live URL.

The deliverable will be a fully accessible web application link submitted with the final report. This URL will be used by the instructor to assess both team and individual contributions to the project tasks, as well as the achievement of course learning outcomes.

Phase 10: Testing and Refinement (Days 26–28)

We will test the application on different browsers and various screen sizes using browser developer tools, as recommended by MDN Web Docs (2024). We will specifically test:

- Responsive design on mobile, tablet, and desktop views.
- User authentication flow (signup, email verification, login, password reset).
- Dashboard functionality (search history, browsing history all working correctly for logged-in users).

- Map interactivity and bi-directional linking with food cards.
- Guest user experience (LocalStorage for favorites and history, merge prompt upon login).
- Review system (create, edit, delete, rating aggregation).
- Cross-browser compatibility (Chrome, Firefox, Safari, Edge).
- Playwright automated screenshot tests for visual regression.

Based on testing, we will refine the user interface and user experience, fix any bugs, and ensure all features work as expected.

Week 4 Deliverable (Final): A fully functional, polished web application deployed to a live URL at <https://food.jackie-macau.top>, with all features working: testable responsive design, full-stack MVC implementation, user authentication (signup/email verification/login/password reset with CSRF protection), user dashboard with search history, browsing history, and favorites, restaurant detail pages with review system, admin panel for content management, interactive map with dual-mode rendering and marker clustering, proper guest user handling, and complete trilingual support.

3.5. Summary Timeline

Week	Focus	Key Deliverables
Week 1	Foundation & User Authentication	Signup/email verification/login/password reset + SQLite database (core relational tables) + 45-50 foods + 15-20 restaurants dataset
Week 2	Front-End & Core Features	Responsive trilingual UI + restaurant/food catalogs + search (debounced) + favorites system (LocalStorage for guests, DB for users)
Week 3	Map, Dashboard & Review System	Interactive map with dual-mode rendering + dashboard (history + favorites) + review system + restaurant detail pages + admin panel
Week 4	Integration, Deployment & Testing	Live URL + fully integrated, tested full-stack web app

4. Criteria for Success

A successful "Taste of Macau" project will be evaluated against several key characteristics aligned with industry standards described by Google (2024) and MDN Web Docs (2024).

Testable Responsive Design – The application must demonstrate fully testable responsive design, meaning the grid of food cards, search bar, filter buttons, embedded map, and user dashboard adapt seamlessly to all screen sizes from large desktop monitors to smartphones. We will use browser developer tools to test and validate responsiveness across multiple device viewports, meeting Google's (2024) core web vitals standards for mobile friendliness.

Visual Appeal – The primary indicator of success will be the visual appeal of the finished product, where the layout is clean, modern, with a cohesive design system (Macau-themed colors: Portuguese tile aesthetics, Chinese red, Portuguese green, egg tart gold), and effectively showcases Macau's culinary heritage through high-quality images and an intuitive interface, as promoted by the Macao Government Tourism Office (2024).

Full-Stack Implementation – The application must demonstrate complete full-stack web development, with front-end (HTML/CSS/JavaScript) providing interactive user interfaces, back-end (PHP) handling server-side logic and authentication, and database (SQLite) persisting user data, search history, browsing history, favorites, and reviews, all connected via MVC architecture.

User Authentication Success – The system must support:

- User signup with email verification (unique link sent via email).
- User login with session management and hashed passwords.
- Password reset flow (forgot password → email reset link → set new password).
- CSRF protection with token generation and single-use rotation.
- All forms must validate user input and provide clear error or success messages.

User Dashboard Success – The user dashboard must be accessible only to logged-in users and must integrate three essential service modules:

1. **Search History** – Displays past search queries made by the user, saved to the database, with ability to re-run searches.
2. **Browsing/Viewing History** – Tracks food cards viewed by the user, persists across sessions, displays "Recently Viewed" items.
3. **Favorites** – Displays dishes saved by the user, synced from guest localStorage to server upon login.

Review System Success – Users can create, edit, and delete reviews with 1–5 star ratings; restaurant rating aggregates (average and distribution) update automatically; duplicate prevention ensures one review per user per restaurant.

Restaurant Detail Pages Success – Each of the 16 restaurants has a dedicated detail page displaying signature dishes, user reviews with star ratings, tag-based attributes, and an embedded map location.

Admin Panel Success – Administrators can perform CRUD operations on restaurants and foods, link foods to restaurants, and manage user feedback reports with a status workflow (new → reviewing → resolved).

Guest User Handling Success – Guest users must have their search and browsing history stored in LocalStorage. Upon login, the system must detect local history and prompt merging with the cloud account. No personally identifiable information should be collected from guest users.

Search Functionality Success – The search feature must correctly filter food items based on user input (by name, category, or location) and update displayed results instantly (client-side). For logged-in users, each search query must be recorded in the database and appear in the dashboard's search history.

Map Interactivity and Clustering Success – Both food markers and restaurant clusters must load correctly. Marker clustering must handle overlapping at lower zoom levels. Bi-directional interaction between food/restaurant cards and map markers must work even when markers are clustered – clicking a card should zoom to reveal and highlight the corresponding marker. The edge case where a restaurant has multiple signature dishes at the same coordinates must be handled correctly.

Trilingual Coverage Success – All content (food entries, restaurant descriptions) and UI labels must render correctly in Chinese, English, and Portuguese, with seamless language switching via AJAX and locale persistence across sessions.

Data Quality and Updateability Success – The initial dataset (49 foods, 16 restaurants) must be verified through the three-layer mechanism. The system must include a user feedback mechanism ('Report Location Issue') and an administrative interface for updating content without code changes.

Successful Deployment with Public URL – The final project must be deployed to a publicly accessible web URL at <https://food.jackie-macau.top> with working SSL. The live site must be accessible, correctly render all front-end content, and have all PHP features (authentication, dashboard, search history, browsing history) functioning as expected. This URL serves as the primary evidence for evaluating team and individual contributions to project tasks and the achievement of course learning outcomes.

Code Quality and Documentation – The HTML must be semantic and accessible, the CSS organized and maintainable, the JavaScript and PHP code well-structured, properly commented, and free of errors according to MDN Web Docs (2024) best practices, following MVC separation of concerns.

Self-Assessment and Testing – To systematically evaluate our final product against these criteria, we will employ a self-assessment checklist based on the characteristics outlined above, incorporating validation standards from W3Schools (2024). Additionally, we will use browser developer tools such as Chrome DevTools for comprehensive responsive testing and debugging, and Playwright for automated screenshot testing, as recommended by MDN Web Docs (2024).

The ultimate proof of success, however, will be the **final visual** and functional **demonstration of the live web application itself**, whose behavior and appearance will serve as the definitive measure of project completion and quality.

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